



Python Programming Fundamentals

Menu-Driven Systems

Programming Fundamentals Team

SETU

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Outline

1. What is a Menu-Driven System?
2. The Basic Menu Loop
3. The match Pattern (Python 3.10+)
4. Getting Valid Input
5. Getting Float Input
6. Menu-Driven System Pattern Summary
7. Full Main Loop Example



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1.1 User-facing control loop

A menu-driven system:

- Presents a numbered list of options
- Reads the user's choice
- Executes the corresponding action
- Repeats until the user quits

This is the **interaction model** for most CLI applications — your Shop app, a student grade system, a contact book, etc.



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2. The Basic Menu Loop

```
def show_menu():  
    print("\n=== Main Menu ===")  
    print("1. Add item")  
    print("2. List items")  
    print("3. Remove item")  
    print("0. Exit")  
  
def main():  
    running = True  
    while running:  
        show_menu()  
        choice = input("Enter choice: ").strip()  
        if choice == "1":  
            add_item()  
        elif choice == "2":  
            list_items()
```



2. The Basic Menu Loop

```
elif choice == "3":  
    remove_item()  
elif choice == "0":  
    running = False  
else:  
    print("Invalid choice – try again.")
```



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3.1 Cleaner than long if/elif chains

```
def main():
    while True:
        show_menu()
        choice = input("Enter choice: ").strip()
        match choice:
            case "1":
                add_item()
            case "2":
                list_items()
            case "3":
                remove_item()
            case "0":
                print("Goodbye!")
                break
            case _:
                print("Invalid choice – try again.")
```



3.1 Cleaner than long if/elif chains

`case _:` is the “default” — it matches anything not caught above.



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4.1 Defensive input reading

```
def get_int_input(prompt: str, min_val: int, max_val: int) -> int:
    while True:
        try:
            value = int(input(prompt))
            if min_val <= value <= max_val:
                return value
            print(f>Please enter a number between {min_val} and {max_val}")
        except ValueError:
            print("Invalid input – please enter a whole number")
```

Usage

```
choice = get_int_input("Choose (0-4): ", 0, 4)
```



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5. Getting Float Input

```
def get_float_input(prompt: str, min_val: float = 0.0) -> float:
    while True:
        try:
            value = float(input(prompt))
            if value > min_val:
                return value
            print(f"Value must be greater than {min_val}")
        except ValueError:
            print("Invalid input – please enter a number")

# Usage in add product:
price = get_float_input("Enter price (>0): ")
```



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6. Menu-Driven System Pattern Summary

Component	Responsibility
<code>show_menu()</code>	Print the numbered options
<code>get_int_input()</code>	Safe validated integer input
<code>while True / running</code>	Keep looping until user quits
<code>match / if-elif</code>	Dispatch to the right function
Action functions	<code>add</code> , <code>list</code> , <code>find</code> , <code>remove</code> , <code>update</code>



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7. Full Main Loop Example

```
from shop import Shop

def main():
    shop = Shop("My Shop")
    running = True
    while running:
        show_menu()
        choice = input("Choice: ").strip()
        match choice:
            case "1": add_product_menu(shop)
            case "2": shop.display_all()
            case "3": find_product_menu(shop)
            case "4": remove_product_menu(shop)
            case "0":
                print("Goodbye!")
                running = False
```



7. Full Main Loop Example

```
        case _:  
            print("Unknown option.")  
  
if __name__ == "__main__":  
    main()
```

Thanks for Watching - Any questions?

